

Multiple Choice

1. **Answer:** A

Options: A) $[(2 e^{-4s})/s^3]$; B) $2 / s^3$; C) $(2 / s^2) * e^{-4s}$; D) $4/s^3$; E) $[(2 e^{-4s})/s^4]$

Note: Correct option: A - $[(2 e^{-4s})/s^3]$

2. **Answer:** E

Options: A) $P(R) = 2/3$, $P(B) = 1/3$; B) $P(R) = 5/14$, $P(B) = 4/14$; C) $P(R) = 1/2$, $P(B) = 1/4$; D) $P(R) = 7/15$, $P(B) = 4/15$; E) $P(R) = 7/16$, $P(B) = 1/4$

Note: Correct option: E - $P(R) = 7/16$, $P(B) = 1/4$

3. **Answer:** A

Options: A) 1 kHz to 100 kHz; B) 500 MHz to 1 GHz; C) 1 Hz to 1 kHz; D) 1 KHz to 1 MHz; E) 10 kHz to 100 MHz

Note: Correct option: A - 1 kHz to 100 kHz

4. **Answer:** A

Options: A) $\{Q^2 / (16\pi\epsilon_0 R)\}$; B) $\{2 Q^2 / (8\pi\epsilon_0 R)\}$; C) $\{Q / (8\pi\epsilon_0 R)\}$; D) $\{Q / (16\pi\epsilon_0 R^2)\}$; E) $\{Q / (4\pi\epsilon_0 R^2)\}$

Note: Correct option: A - $\{Q^2 / (16\pi\epsilon_0 R)\}$

5. **Answer:** B

Options: A) 400 db/meter; B) 545 db/meter; C) 600 db/meter; D) 325 db/meter; E) 250 db/meter

Note: Correct option: B - 545 db/meter

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Off Switch'.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: ' $(1 / 4)(1 - \cos 2 t)$ '.

9. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'it is a square wave signal'.

10. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

Long Form Response

11. **Answer:** 1160 ft/sec. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** 1.9 ohms. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key